

multiplication

In decimal long multiplication, we multiply by one digit at a time, shift each row one place further left, and add the rows.

$$\begin{array}{r} 12 \\ \times 13 \\ \hline 36 \quad 12 \times 3 \\ + 120 \quad 12 \times 1, \text{ shifted one place} \\ \hline 156 \end{array}$$

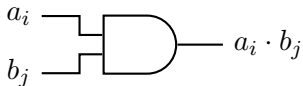
binary works the same way, with much simpler rows

Every binary digit is 0 or 1, so each row is either a copy of the first number or all zeros.

$$\begin{array}{|c|c|c|c|} \hline 1 & 0 & 1 & 1 \\ \hline \end{array} \times 1 = \begin{array}{|c|c|c|c|} \hline 1 & 0 & 1 & 1 \\ \hline \end{array}$$

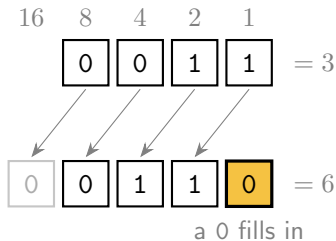
$$\begin{array}{|c|c|c|c|} \hline 1 & 0 & 1 & 1 \\ \hline \end{array} \times 0 = \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array}$$

Each bit of a row is 1 only when both bits are 1: an **AND**.



one gate per bit
builds every row at once

Shifting every bit one place left moves it to a weight twice as large, so the value doubles.



shifting left by k places multiplies by 2^k

To multiply 1011 (11) by 1101 (13), we write one row for each bit of the second number, shifted left by that bit's position.

			1	0	1	1		
		×	1	1	0	1		
			1	0	1	1		bit 0 is 1: copy
+			0	0	0	0		bit 1 is 0: zeros
+		1	0	1	1			bit 2 is 1: copy, shifted 2
+	1	0	1	1				bit 3 is 1: copy, shifted 3

Then we add the rows, one column at a time, carrying as usual.

$$\begin{array}{r}
 \begin{array}{cccc}
 & & 1 & 0 & 1 & 1 \\
 & \times & 1 & 1 & 0 & 1 \\
 \hline
 & & 1 & 1 & 1 & 1 & & \\
 & & 1 & 0 & 1 & 1 & & 11 \\
 + & & 0 & 0 & 0 & 0 & & 0 \\
 + & & 1 & 0 & 1 & 1 & & 44 \\
 + & 1 & 0 & 1 & 1 & & & 88 \\
 \hline
 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 = 143 = 11 \times 13
 \end{array}
 \end{array}$$

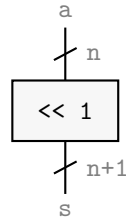
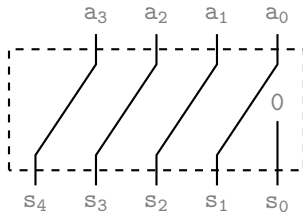
The product of two n -bit numbers can need $2n$ bits. Even $15 \times 15 = 225$ needs eight.

$$\boxed{n \text{ bits}} \times \boxed{n \text{ bits}} = \boxed{2n \text{ bits}}$$

$$15 \times 15 = 225 = \underbrace{\begin{array}{|c|c|c|c|} \hline 1 & 1 & 1 & 0 \\ \hline \end{array}}_{\text{lost if we keep only 4 bits}} \underbrace{\begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 1 \\ \hline \end{array}}_{0001 = 1, \text{ not } 225}$$

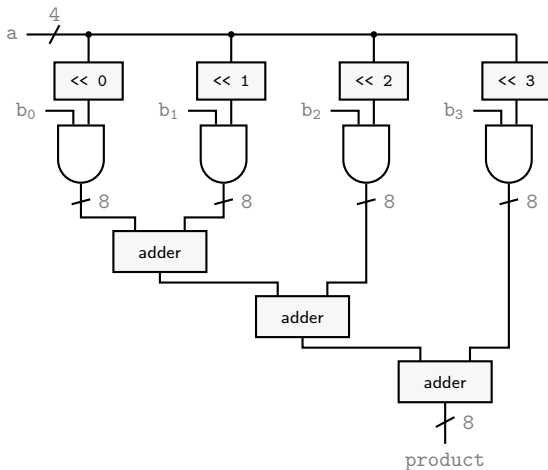
$$(2^n - 1)^2 < 2^{2n}, \text{ so } 2n \text{ bits always suffice}$$

A **shifter** moves every bit one place to the left. A fixed shift needs no gates at all, only wiring.



a shift by k places is the same wiring, moved over k places

In hardware, each row is a passed through a shifter and ANDed with one bit of b . A chain of adders sums the rows.



each adder is the chain of full-adders from before

Following 1011×1101 through the circuit, one row at a time:

row	b_i	shifter	AND	running sum
0	1	00001011	00001011	00001011 11
1	0	00010110	00000000	00001011 11
2	1	00101100	00101100	00110111 55
3	1	01011000	01011000	10001111 143

a 0 bit zeroes its row; the last adder gives the product

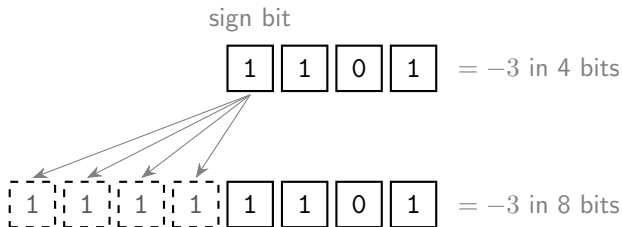
In two's complement, the top bit weighs a *negative* amount. So when the second number is negative, its last row is subtracted, not added.

$$\begin{array}{|c|} \hline 1 \\ \hline \end{array} \begin{array}{|c|} \hline 1 \\ \hline \end{array} \begin{array}{|c|} \hline 0 \\ \hline \end{array} \begin{array}{|c|} \hline 1 \\ \hline \end{array} = -3$$

$-8 \quad 4 \quad 2 \quad 1$

$$A \times 1101 = A \cdot 1 + A \cdot 4 - A \cdot 8 = A \cdot (-3)$$

The rows are now $2n$ bits wide. To widen a negative number without changing its value, we copy its sign bit into the new places.



filling with zeros instead would give 00001101, which is 13

$(-3) \times 5$: the rows are copies of 1101, sign-extended to eight bits.

Diagram illustrating the multiplication of -3 and 5 using bit shifts and addition.

Initial values:

- -3 (binary: 1101)
- 5 (binary: 0101)

Multiplication process:

- Step 1: 1×1101 (bit 0: copy -3)
- Step 2: 0×1101 (bit 1: zeros)
- Step 3: 1×1101 (bit 2: shifted 2 (-12))
- Step 4: 0×1101 (bit 3: zeros)

Final result: -15 (binary: 1111)

$5 \times (-3)$: the top bit of 1101 weighs -8 , so the last row adds -5 , shifted three places.

				-8					
				0	1	0	1	5	
			$\times$	1	1	0	1	-3	
				0	0	0	0	0	1
						$\overset{1}{\uparrow}$			
						0	1	0	1
+						0	0	0	0
									bit 1: zeros
+				0	0	0	1	0	1
									bit 2: shifted 2 (20)
+				1	1	0	1	1	
									bit 3: -5 , shifted 3 (-40)
				1	1	1	1	0	0
								0	1
									$= -15$

try it

step through a multiplication of your own